

Econometric_HW0317_314707033 蔡霽沅

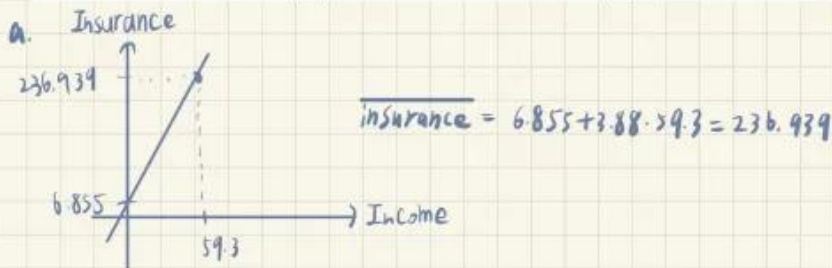
#3.18

3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let *INCOME* be household income (thousands of dollars) and *INSURANCE* the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of *INCOME* = 59.3. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.



b. ① point estimate

$b_2 = 3.88$, meaning insurance increase by about
\$3.88 for each additional \$1,000 of income.

② 95% C.I. $df = 20 - 2 = 18$

$$[b_2 - t_{0.025}(18) \cdot se(b_2), b_2 + t_{0.025}(18) \cdot se(b_2)]$$

$$= [3.88 - 2.101 \cdot 0.112, 3.88 + 2.101 \cdot 0.112] = [3.645, 4.115]$$

The 95% C.I. is $[3.645, 4.115]$, indicating we are 95%
confident that with each additional \$1,000 of income,
the insurance held by household will be within this range.

- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .

$$C. \quad 100,000 \div 1,000 = 100$$

$$\widehat{insurance} = 6.855 + 3.88 \cdot 100 = 394.855$$

$$\begin{aligned} \text{Var}(\widehat{insurance}) &= \text{Var}(b_1) + 100 \cdot \text{Var}(b_2) + 2 \cdot 100 \cdot \text{Cov}(b_1, b_2) \\ &= 7.383^2 + 100^2 \cdot 0.112^2 + 200 \cdot (-0.746) \\ &= 54.509 + 125.44 - 149.2 = 30.749 \end{aligned}$$

$$SE(\widehat{insurance}) = \sqrt{30.749} = 5.545$$

$$99\% \text{ C.I.}, df = 20 - 2 = 18 \quad \widehat{insurance} \pm t_{0.005}(18) \cdot SE(\widehat{insurance})$$

$$\Rightarrow [394.855 \pm 2.878 \cdot 5.545] = [378.896, 410.814]_{**}$$

The 99% C.I. for the expected insurance at \$100,000 income is $[378.896, 410.814]_{**}$

- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.
- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

$$D. \quad \begin{cases} H_0: b_2 = 5 \\ H_a: b_2 \neq 5 \end{cases} \quad \textcircled{2} \alpha = 0.05 \quad \textcircled{3} T = \frac{b_2 - 5}{SE(b_2)} \stackrel{H_0}{\sim} t_{(20-2)}$$

$$\textcircled{4} RR = \{|T| \geq t_{0.025}(18) = 2.101\} \quad \textcircled{5} T_0 = \frac{3.88 - 5}{0.112} = -10$$

$\textcircled{6} \because T_0 \in RR, \therefore$ reject H_0 , the data do not support the claim.

$$E. \quad \begin{cases} H_0: b_2 = 1 \\ H_a: b_2 > 1 \end{cases} \quad \textcircled{2} \alpha = 0.01 \quad \textcircled{3} T = \frac{b_2 - 1}{SE(b_2)} \stackrel{H_0}{\sim} t_{(20-2)}$$

$$\textcircled{4} RR = \{T \geq t_{0.01}(18) = 2.552\} \quad \textcircled{5} T_0 = \frac{3.88 - 1}{0.112} = 25.714$$

$\textcircled{6} \because T_0 \in RR, \therefore$ reject H_0 , the slope is significantly not equal to 1.

#3.23

(a)

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	Item	Value

1	t-statistic	-26.729
2	Critical value	1.648
3	P-value	1

Since the t-statistic doesn't exceeds the critical value and the p-value doesn't less than 0.05, we **don't reject** the null hypothesis. There is no evidence that the marginal effect is greater than \$13,000.

(b)

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	Item	Value

1	t-statistic	4.189
2	Critical value	1.648
3	P-value	0.00002

Since the t-statistic exceeds the critical value and the p-value less than 0.05, we **reject** the null hypothesis. There is sufficient evidence to conclude that the marginal effect is significant greater than \$13,000.

(c)

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	Interval	Value

1	LowerBound	158.048
2	UpperBound	176.699

In repeated sampling, about 95% intervals constructed this way will contain the true value of the average price of a 2000 square foot house.

(d)

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	Mean	Value

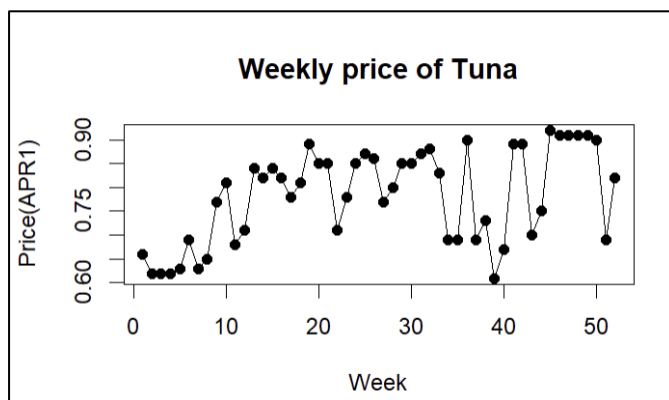
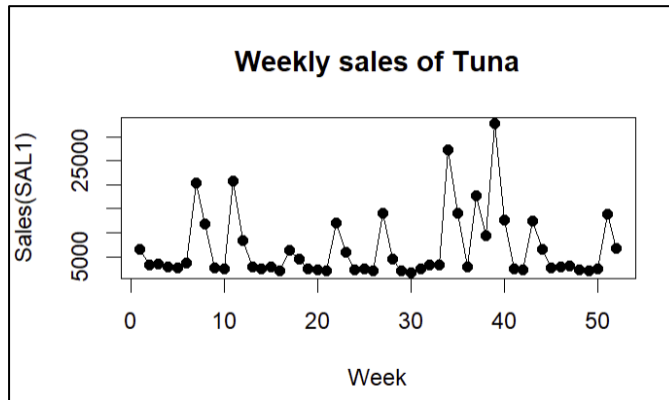
1	price	163.333

The sample mean price for houses with 2000 square feet falls within the 95% confidence interval. Therefore, the sample average of the selling price for houses with SQFT = 20 is compatible with the result in part (c).

#3.31

(a)

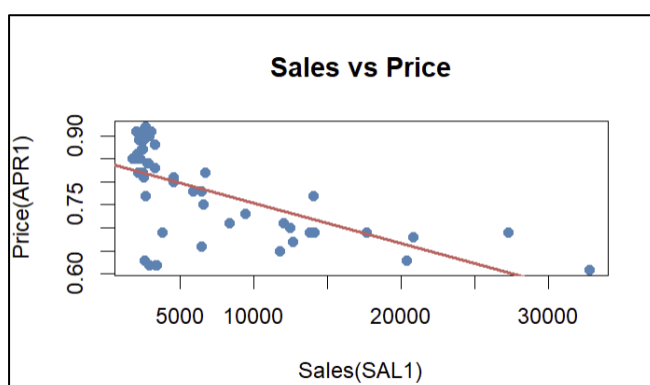
Statistic	Mean	Min	Max	St. Dev.
sal1	6,718.712	1,762	32,820	6,915.083
apr1	0.782	0.610	0.920	0.098



There is substantial variation in both sales and price from week to week. Sales fluctuate considerably, with several sharp spikes and drops, while prices also vary over time, although less dramatically. Overall, both series show frequent fluctuations.

Moreover, the plots suggest a negative relationship between price and sales. When the price increases, sales tend to decrease, and when the price decreases, sales tend to increase.

(b)



The plot shows a negative relationship between sales and price, indicating that sales tend to decrease as price increases. The result is as expected and is consistent with the law of demand.

(c)

Dependent variable:	

sal1	

price1	-434.447*** (78.585)
Constant	40,714.220*** (6,196.421)

The point estimate for the effect of a one cent increase in the price on the sales is equal to -434.447.

Interval		Value
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1	LowerBound	-592.290
2	UpperBound	-276.605
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(d)

Interval		Value
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1	LowerBound	8,652.954
2	UpperBound	11,952.850
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(e)

Type		Value
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price1	price elasticity	-5.060
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Interval		Value
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1	LowerBound	-6.858
2	UpperBound	-3.262
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The absolute value of the price elasticity is greater than 1, indicating that demand is highly elastic. This makes economic sense, as there are many brands of tuna cans that consumers can easily switch to as substitutes.

(f)

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	Item	Value

1	t-statistic	-2.251
2	Critical value	1.676
3	P-value	0.029

Since the t-statistic exceeds the critical value and the p-value less than 0.1, we reject the null hypothesis. Therefore, there is sufficient evidence to conclude that the price elasticity of sales for brand no.1 canned tuna is significantly different from -3.